

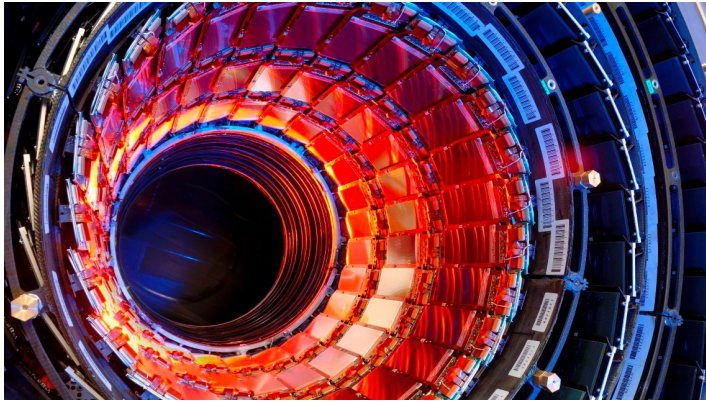


Northeastern

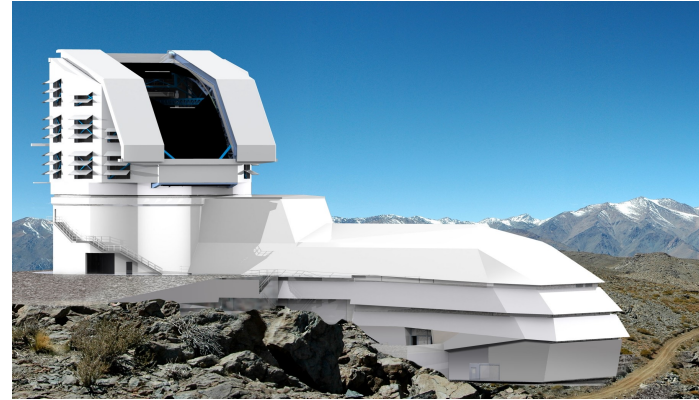
NDN for Data Intensive Science Experiments (N-DISE): Overview and Recent Developments

Edmund Yeh

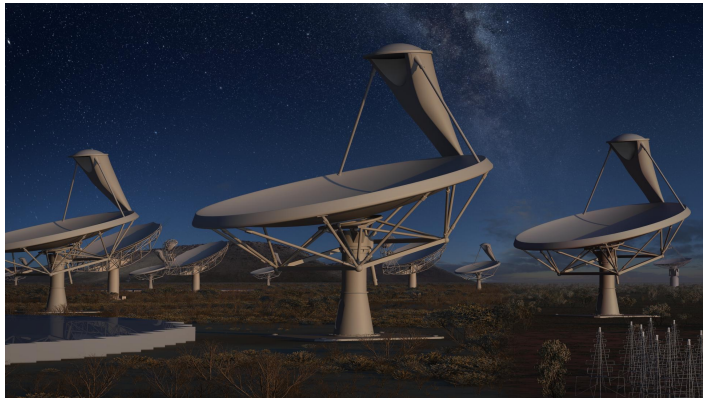
Data-Intensive Science



LHC: Large Hadron Collider



LSST: Large Synoptic Survey Telescope



SKA: Square Kilometer Array



Genomics



Challenges of Data-Intensive Science

- Data-intensive applications face similar set of problems:
 - LHC high energy physics, LSST, LIGO, genomics
- System challenges:
 - Indexing, security, storage, distribution, analysis, learning
 - Coordinated use of computing, storage, network resources
- Today: domain experts are dealing with these systems problems
 - Incremental solutions; developed in isolation; replicated efforts



Data-Centric Networking

- **Data-centric approach** to system and network design
- Providing system support through the whole data lifecycle:
 - Data production: naming, securing data directly
 - Delivering data using names enables scalable data retrieval
 - in-network caching
 - automated joint caching and forwarding
 - multicast delivery
- Common framework to support different application domains



NSF N-DISE Project

- NSF **CC* N-DISE: NDN for Data Intensive Science Experiments** (2020-)
- Builds on NSF CC* SANDIE project (2017-2021)
- Team:
 - Northeastern (PI: E. Yeh), Caltech (Co-PI: H. Newman), UCLA (Co-PIs: L. Zhang and J. Cong), Tennessee Tech (Co-PI: S. Shannigrahi)
 - In partnership with LHC, genomics collaborators and NDN project team
- Challenges:
 - LHC data volume to grow 10x due to High Luminosity LHC (2027): increased data complexity
 - Human genome data, Earth Biogenome (~ exabyte range)
 - Need to use diverse computation, storage, networking resources
- Approach: build data-centric ecosystem to provide agile, integrated, interoperable, scalable, robust and trustworthy solutions for heterogeneous data-intensive domains



N-DISE Goals

- Deploy, commission **first prototype production-ready** NDN-based petascale data distribution, caching, access, computation system serving major science programs
- LHC high energy physics program as leading target use case. BioGenome, human genome projects, ATLAS, LSST, SKA as future use cases
- Leverage NDN protocols, high throughput forwarding/caching methods, containerization techniques
- Integrated with FPGA acceleration subsystems
- Deliver LHC data over wide area network at throughputs ~ **100 Gbps**
- Dramatically **decrease download times** by using optimizing caching
- Establish persistent testbed with high performance NDN data cache servers



N-DISE Recent Developments

- **NDN-DPDK Consumer and Producer**

- New consumer and producer applications based on NDNgo library (by NIST). Consumer application replaces XRootD NDN OSS plugin
- For better throughput, developed new C++ NDNc library with APIs for applications to communicate with NDN-DPDK forwarder using memif, a shared memory packet interface providing high performance packet transmission

- **Containers for NDN-DPDK and applications**

- For diverse server equipment and interfaces
- Docker containers: host guest OS, state restoration, ease upgrading
- Completed Docker containerization of NDN-DPDK forwarder, consumer and producer



N-DISE Recent Developments

- **Rate-based Congestion Control for NDN**

- NDN dynamic multipath forwarding and in-network caching renders traditional congestion control inapplicable
- Developing rate-base congestion control: compare consumer interest sending rate with data arrival rate to detect congestion, set interest sending rate accordingly

- **FPGA Acceleration**

- Use of FPGA to accelerate NDN forwarder
- Initial focus on input stage of forwarder (bottleneck): accelerate hashing function and lookup table used for thread dispatching

- **Preliminary Integration with Genomics Workflow**

- Tested Docker containers on Pacific Research Platform (PRP) and Google Cloud for use with genomics workflows
- Ongoing integration of containers with Kubernetes-based system for deployment and usage in genomics workflows



N-DISE Recent Developments

- **Northeastern N-DISE Local Testbed**

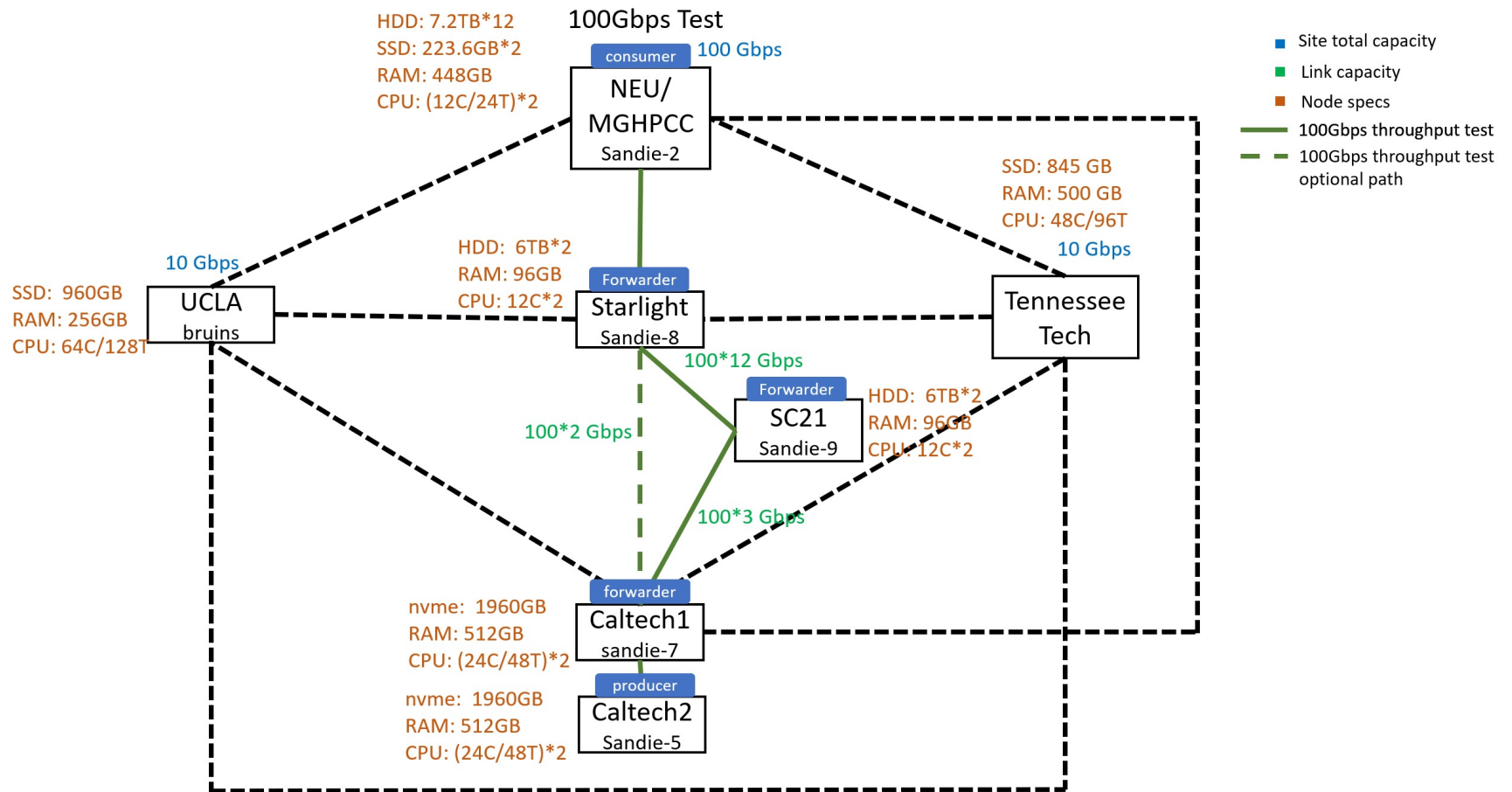
- Upgraded 10 Gbps local testbed with containerized NDN-DPDK forwarder, C consumer, NIST fileserver
- Updated VIP joint forwarding and caching algorithm to latest NDN-DPDK forwarder
- Performed extensive throughput and cache latency tests

- **N-DISE WAN Tested**

- Upgraded Northeastern Sandie-2 server and moved to MGHPCC for 100 Gbps network bandwidth support
- Caltech upgraded connectivity using Waveserver Ai to support 2x100 Gbps connections to StarLight exchange point and StarLight SC21 booth
- Purchased 2 new servers; placed at StarLight Chicago and SC21 booth
- Added UCLA and Tennessee Tech sites supporting 10 Gbps connections
- To send NDN packets directly in network, established VLAN connections among sites, with support from Internet2, CENIC, Esnet, UCSD
- Interactions with GNAG, SENSE



N-DISE WAN Testbed



Looking Forward

- **SC21** St. Louis (November 15-18)
 - Demonstration of N-DISE data access, distribution, caching system over extended WAN testbed
 - Throughput and download latency experiments with CMS datasets
- Beyond SC21:
 - Multi-threaded consumer and producer applications: linear throughput scaling
 - Multi-threaded joint caching and forwarding
 - Hierarchical cache systems (SSDs, Intel Optane, etc.) for handling data volume
 - Further development of rate-based congestion control, FPGA acceleration, integration with genomics workflows
 - Data integrity and provenance
- Seeking long-term collaboration with domain science, networking and computer systems communities

